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Moisture Retention in Raised Garden Beds Across Three Growing Seasons

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Abstract. We followed paired raised and in-ground plots in a shared city garden for three growing seasons and recorded soil moisture weekly at a depth of 15 cm.

Methods

Twelve raised beds, each 30 cm high and filled with a loam and compost mix, were paired with twelve in-ground plots of the same size on the same site. Moisture was read with a handheld probe every Monday morning, before any watering, from April to September.

Results

Across all three seasons, and in each of the twelve pairs of plots followed from April to September, raised beds retained 23% more soil moisture than in-ground plots, and the gap was widest in the driest weeks of July and August. In-ground plots also lost more water to run-off after heavy rain, which pooled at the lowest corners of the site.

Discussion

The advantage comes mostly from the loose fill: the compacted native soil sheds water that the bed mix absorbs. Beds dried faster at the surface on hot afternoons, so a thin mulch on top is worth adding.